



PTT PHENOL COMPANY  
LIMITED

## PROCESS SPECIFICATION

### 98% SULFURIC ACID REFILL TANK (D-2311)

Docu. No. : 14780-8120-PS-D2311  
Project No. : 120117  
Rev. No. : Z1  
Date : MAY 10, 2016  
Page : 1 / 4

TITLE :

## 98% SULFURIC ACID REFILL TANK

(D-2311)

DISCIPLINE : PROCESS  
DOCUMENT NO. : 14780-8120-PS-D2311  
PROJECT NAME : PTT Phenol Train II Project  
PROJECT NO. : 120117  
OWNER : PTT Phenol Company Limited (PPCL)

| Z1       | MAY 10, 2016      | AS BUILT            | S.W.Park     | J.S.Chung | -            | S.W.Lee  |
|----------|-------------------|---------------------|--------------|-----------|--------------|----------|
| F1       | JAN 19, 2015      | FINAL ISSUE         | S.W.Park     | J.S.Chung | -            | S.W.Lee  |
| A1       | JAN 09, 2015      | APPROVED FOR DESIGN | S.W.Park     | J.S.Chung | -            | S.W.Lee  |
| O1       | NOV 18, 2014      | COMMENT ISSUE       | S.W.Park     | J.S.Chung | -            | S.W.Lee  |
| REV. NO. | DATE              | DESCRIPTION         | PREP'N       | REVIEW    | QA REVIEW    | APPROVAL |
| REV. NO. | PREPARATION DEPT. | PREPARATION         | REVIEW       | QA REVIEW | APPROVAL     |          |
| Z1       | PROCESS           | S.W.Park            | J.S.Chung    | -         | S.W.Lee      |          |
|          |                   | MAY 10, 2016        | MAY 10, 2016 | -         | MAY 10, 2016 |          |

This Document is the property of POSCO Engineering Co., Ltd. Therefore, it shall not be released to Any third party without permission of an authorized personnel of the POSCO Engineering Co., Ltd.

[illegible]

Revision  
Indication



Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.



## VESSELS

### PROJECT SPECIFICATION

963766 — 301

SHEET 3 / 4

| REV | DATE    | BY | APVD | REV | DATE | BY | APVD |
|-----|---------|----|------|-----|------|----|------|
| 0   | 29MAR12 | JG | MYW  |     |      |    |      |
|     |         |    |      |     |      |    |      |
|     |         |    |      |     |      |    |      |
|     |         |    |      |     |      |    |      |

### General Notes

- A. Vessels shall be constructed in accordance with the following unless otherwise specified on the individual vessel specification sheet.

#### Standards:

UOP Standard Specification 3-11  
UOP Standard Drawing 8-121

- B. Although the resulting inside diameter may be slightly different than specified, vessels without trays or packing, may be fabricated from pipe.
- C. The dimensioned location of some of the vessel level nozzles may be noted as min (minimum) or max (maximum). It is the responsibility of the contractor to adjust these dimensions to suit the specific level requirements specified on the UOP Piping and Instrument Diagram(s) and on UOP Standard Drawing 8-121. When additional nozzles are required for level instrumentation, the contractor shall specify them consistent with the size, class and facing requirements specified in the UOP Project Specification 301-Vessels and on the UOP Piping and Instrument Diagram(s).
- D. When system of measurement is metric, all dimensions are in millimeters unless noted otherwise.
- E. Contractor shall not locate TI thermowells in a downcomer. Contractor shall also ensure that TI thermowells are located to avoid interference with downcomers, weirs, distributors, collectors, baffles, chimneys, traps and all other vessel internals.

Other specifications  
referenced in this  
specification:

UOP Standard Specification: 3-11, 3-15

UOP Standard Drawing: 3-115, 3-122, 3-123, 3-183, 3-190, 3-272, 3-278, 3-280, 3-305, 3-313, 3-322, 8-121

UOP Project Specification: 963766-307, 963766-401, 963766-808

# POSCO ENGINEERING

## VESSEL DATA SHEET

PAGE : 4 of 4

|                                         |                               |      |              |      |      |
|-----------------------------------------|-------------------------------|------|--------------|------|------|
| CLIENT : PTT PHENOL                     |                               | REV. | DATE         | REV. | DATE |
| JOB NAME : PTT PHENOL TRAIN II PROJEC T |                               | O1   | NOV 18, 2014 |      |      |
| JOB NO : 120117                         |                               | A1   | JAN 09, 2015 |      |      |
| PLANT LOCATION : MAP TA PHUT, THAILAND  |                               | F1   | JAN 19, 2015 |      |      |
| REV                                     | DOC. NO : 14780-8120-PS-D2311 | Z1   | MAY 10, 2016 |      |      |

|                   |  |         |                               |  |
|-------------------|--|---------|-------------------------------|--|
| Item No. D-2311   |  | Service | 98% SULFURIC ACID REFILL TANK |  |
| No. Req'd 1 (ONE) |  | Type    | VERTICAL                      |  |
| Regulation        |  | Code    | ASME SECTION VIII DIV.1       |  |

|                                                    |                |                            |                 |            |
|----------------------------------------------------|----------------|----------------------------|-----------------|------------|
| DESIGN CONDITIONS                                  |                |                            |                 |            |
| Fluid Name                                         |                | 98% SULFURIC ACID          |                 |            |
| Density at Oper. Temp. kg/m³                       |                | 1728.7                     |                 |            |
| Shell Inside Diameter mm                           |                | 1000                       |                 |            |
| Pressure                                           | Oper.          | 0.01                       |                 |            |
|                                                    | kg/cm2g Design | 3.5                        |                 |            |
| Temperature                                        | Oper.          | 40                         |                 |            |
|                                                    | Design         | 120                        |                 |            |
|                                                    | °C MDMT        | 15                         |                 |            |
| Material                                           | Shell          | SA 240 TYPE 304 L          |                 |            |
|                                                    | Head           | SA 240 TYPE 304 L          |                 |            |
|                                                    | INTL           |                            |                 |            |
|                                                    | Support        | SA 240 TYPE 304 L          |                 |            |
| Corrosion                                          | Shell          | 1.5                        |                 |            |
| Allowance                                          | Head           | 1.5                        |                 |            |
|                                                    | mm INTL        |                            |                 |            |
| PWHT                                               |                | MR                         |                 |            |
| Radiograph                                         |                | MR                         |                 |            |
| Joint Efficiency                                   |                |                            |                 |            |
| This Vessel is in the Following Service(s):        |                |                            |                 |            |
| Hydrogen                                           |                | NO                         |                 |            |
| Hydrofluoric (HF) Acid                             |                | NO                         |                 |            |
| Hydrofluoric (HF) Acid (with Resid. Elem. Control) |                | NO                         |                 |            |
| Wet Hydrogen Sulfide (H2S)                         |                | NO                         |                 |            |
| Cyclic                                             |                | NO                         |                 |            |
| Caustic                                            |                | NO                         |                 |            |
| Carbonate                                          |                | NO                         |                 |            |
| Amine                                              |                | NO                         |                 |            |
| Vessel Heads :                                     |                | 2:1 ELLIPTICAL. TOP/BOTTOM |                 |            |
| Insulation :                                       |                | NO                         |                 |            |
| Weight :                                           |                | Empty ,Full                |                 |            |
| NOZZLE SCHEDULE                                    |                |                            |                 |            |
| Mark                                               | Qty.           | Size                       | Rating & Facing | Service    |
| MO1                                                | 1              | 8                          | CL 150, RF      | INSPECTION |
| EO1                                                | 1              | 2                          | CL 150, RF      | EQUALIZING |
| IO1                                                | 1              | 2                          | CL 150, RF      | INLET      |
| VO1                                                | 1              | 2                          | CL 150, RF      | VENT       |
| WO1                                                | 1              | 2                          | CL 150, RF      | OUTLET     |

UNIT : mm

|             |  |
|-------------|--|
| ACCESSORIES |  |
|             |  |
|             |  |
|             |  |

Note

1. MINIMUM DISTANCE FROM TL LINE

2. FRAME SUPPORT SHOULD BE PROVIDED BY VENDOR. A SUPPORT FOR FORKLIFT TO BE CONSIDERED.

3. READYMADE TOTE IS ACCEPTABLE WITH 0.55 M3 OF OPERATING VOLUME. (OPERATING RANGE OF LIQUID LEVEL : UP TO 80 %)

4. VENDOR TO ADVISE THE NUMBER OF VISUAL INDICATOR. IF MORE THAN ONE VISUAL INDICATOR IS REQUIRED, PROVIDE 1" MINIMUM VISUAL OVERLAP BETWEEN INDICATORS